

EEE 443/543 FALL 2025-26

Assignment # 1

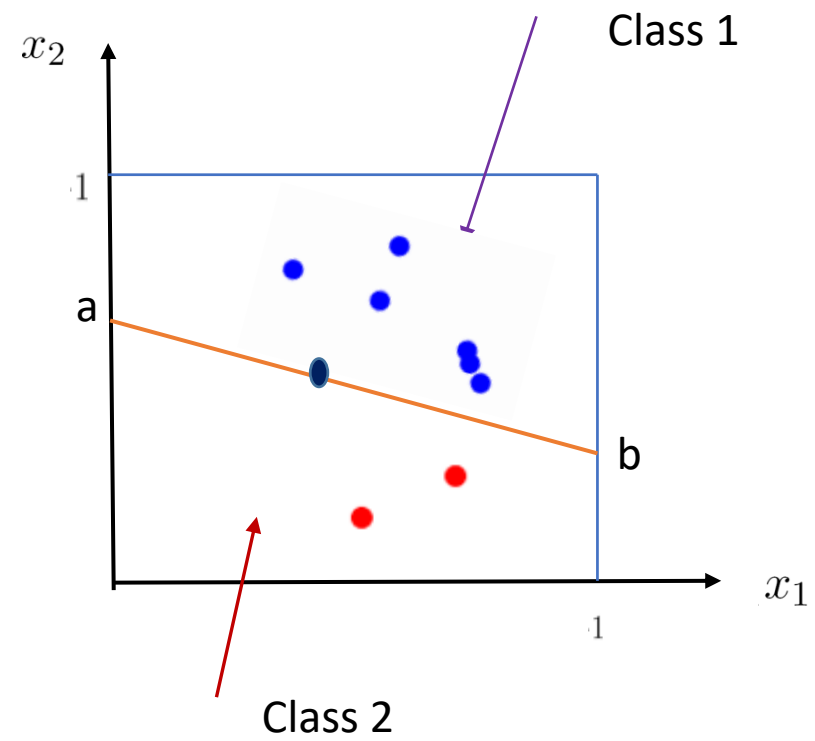
Assignment : Write a computer program (in python XXXXXXXXXX) that implements the perceptron training algorithm for 2 class case with the step activation function. Implement the following steps and report your results. Upload your code (which should be executable by the evaluators) that provides these results as well as your final report separately.

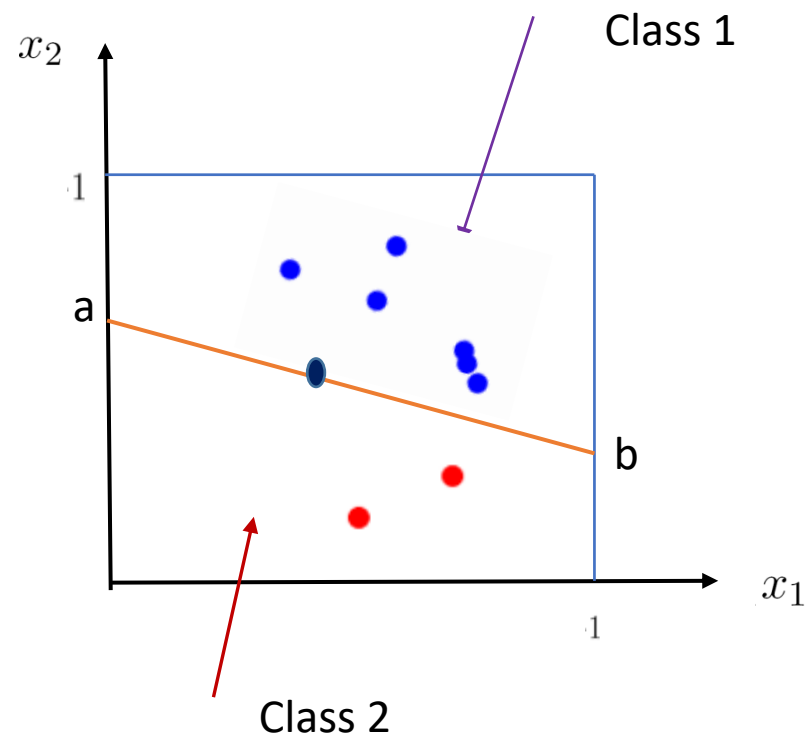
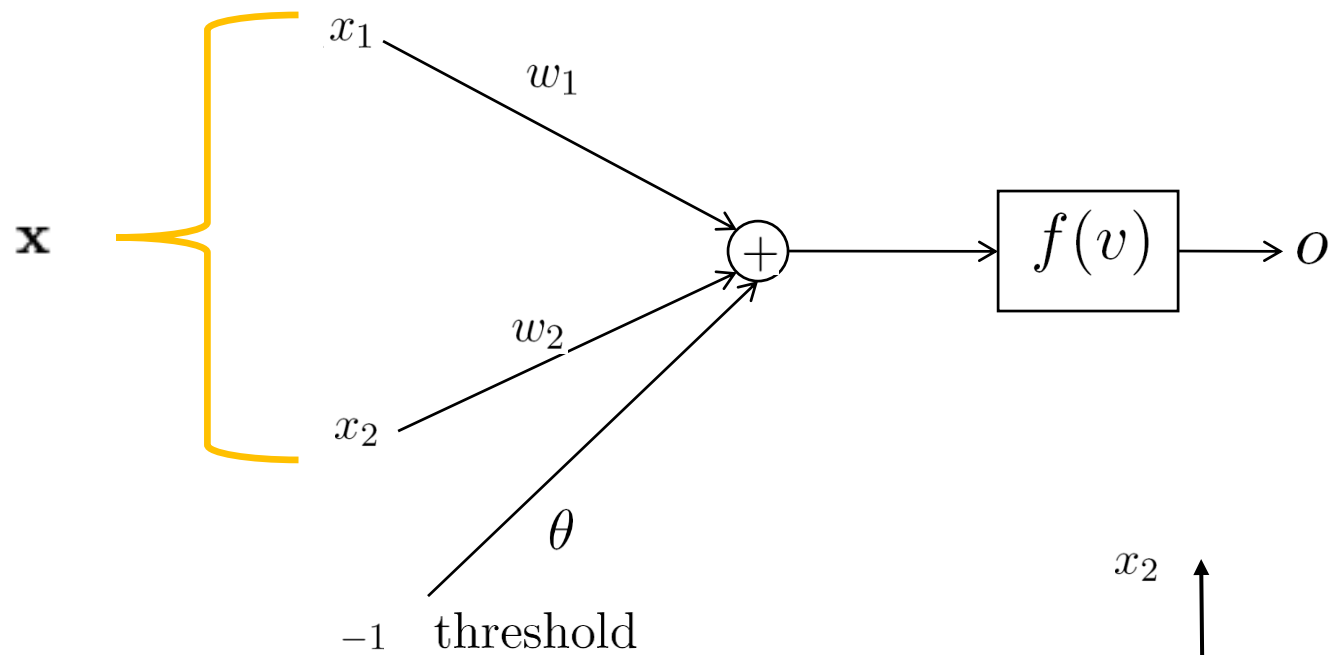
1 : Let $x = (x_1 \ x_2)^T \in \mathbf{R}^2$. Select $n = 100$ points $\mathbf{x}_1, \dots, \mathbf{x}_n$ randomly in unit square, i.e. $x_i \in [0, 1]$. (This should be handled by your code). Call these collection of points as $\mathcal{S}$.

2 : Select two numbers $a, b \in [0, 1]$ randomly. Consider the line which passes between the points $(0, a)$ and $(1, b)$ in the $x_1 - x_2$ plane. Call the class 1, $\mathcal{C}_1 \subset \mathcal{S}$ which are above or on of this line and class 2 as the other points.

3 : Plot these points with suitable labels (e.g. red dots for class 1, blue dots for class 2, etc.) as well as the separating line given above.

4 : We will use a single perceptron with two inputs x_1, x_2 , one threshold and one output, where desired output will be 1 for class 1 and 0 for class 2.





The following steps 5-7 should be repeated for learning coefficient $\eta = 0.1, 1, 10$.

5 : Choose the initial weights w_1, w_2, θ randomly on $[-1,1]$. With these weights find the number of misclassified points. Run the perceptron training algorithm until convergence. At each epoch, find the number of misclassified points.

6 : Repeat step 3, draw the separating line obtained in step 5 as well as the original separating line given in step 2. Make comments on the similarities, differences, etc., if any.

7 : Plot the epoch number vs. misclassification number. Make reasonable comments if you see any patterns related to the convergence.

8 : Compare the results you obtained for different η values.

9 : Repeat steps 1-8 for $n = 500$ patterns.

10 : Write your results in a nice report format, preferably in pdf.

Instructions:

1. Prepare a report (including your answers/plots) to be uploaded on Moodle.
 2. The report should be typeset (no handwriting allowed except for lengthy derivations, which may be scanned and embedded into the report).
 3. Show all steps of your work clearly.
 4. Unclear presentation of results will be penalized heavily.
 5. No partial credits for unjustified answers.
 6. **Use of any toolbox or library for neural networks is prohibited.**
 7. Return all XXXXXXXXXX Python code that you wrote in a single XXXX.py file.
 8. Code should be commented,
 9. The code file should NOT return an error during runtime.
 10. If the code returns an error at any point, the remaining part of your code will not be evaluated (i.e., 0 points).
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